
DisorderFlow

Bayesian Flow Networks for Disorder-Aware Antibody CDR-H3 Design

Team 28 — Analyzing Disorder

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1. The IDP Antibody Design Challenge

- Intrinsically disordered proteins (IDPs) lack stable 3D structure yet drive neurodegenerative diseases (Alzheimer's, Parkinson's)
- Therapeutic targets: Amyloid-beta, tau, alpha-synuclein — all disordered
- Most antibody design methods assume rigid, well-ordered epitopes — fundamentally inadequate for IDP targets
- CDR-H3 is the primary determinant of antigen specificity and the most diverse loop in the antibody structure
- Key gap: no computational framework couples epitope disorder profiles to CDR-H3 sequence selection

Goal: Build a disorder-aware scoring framework that conditions CDR-H3 design on both backbone geometry and per-residue epitope disorder profiles.

2. DisorderFlow Platform Architecture

BFN Core

Categorical distributions over
20 amino acids; iterative
refinement via receiver net

Disorder Head

Per-residue disorder prediction;
focal-weighted loss ($\gamma=4.0$);
handles 0.74% class imbalance

Position Routing

Direct coupling of antigen
residue disorder values to
CDR-antigen pair features

Contrastive Loss

Factual vs shuffled/mismatched
disorder profiles; rank-aware
training signal

Training Pipeline

SAbDab2 structural database | 6-axis homology-disjoint split (PDB, paired Ab, VH, VL, H3, antigen)

Backbone coordinates + disorder labels + negative profiles | ProteinMPNN v_48_020 for evaluation

3. ECLS: Epitope-Conditioned Likelihood Shift

$$\text{ECLS}(s) = \text{NLL_complex}(s) - \text{NLL_peptide-stripped}(s)$$

Measures whether a CDR-H3 sequence is preferentially supported by the presence of peptide epitope coordinates. Positive advantage = native sequence benefits more from epitope context than composition-matched shuffles (200 per record).

Methodology

- Score only heavy chain; fix non-H3 positions; retain light chain + peptide context
- 200 composition-matched shuffles per record as counterfactual controls
- Official SAbDab2 CDR-H3 annotations; heavy-atom distance $\leq 4.5 \text{ \AA}$ contact gate
- Official antigen cluster as inference unit; 10,000 bootstrap resamples for CIs
- One-shot evaluation on sealed temporal data; frozen gates set before evaluation

4. Results: ECLS Detection and Temporal Transfer

Adaptation Set (46 clusters)

Metric	Value
Mean ECLS advantage	0.217
Median	0.203
95% Bootstrap CI	[0.146, 0.290]
Positive clusters	80.4%

Temporal Final (15 clusters, post-2021)

Metric	Value
Mean ECLS advantage	0.172
Median	0.119
95% Bootstrap CI	[0.059, 0.292]
Positive clusters	80.0%

Both frozen gate sets passed. Deposited CDR-H3 sequences carry measurable epitope-conformation-specific signal that transfers to post-training structures.

Sign-flip $P < 1e-6$ (adaptation) and $P = 0.014$ (temporal final).

Key: 80%+ positive clusters on sealed temporal data confirms non-trivial epitope-H3 coupling

5. T2.1 v2: Deterministic Torsion Recovery Benchmark

Deterministic phi/psi torsion perturbation to 1.5–3.0 Å RMSD tier, then test if native residue contacts guide recovery. Peptide restraint: 100 kJ/mol/nm².

Metric	Value
Total structures	31
Torsion hits (target tier)	26/31 (83.9%)
Valid structures after recovery	25/26 (96.2%)
Mean held-out contact recovery	0.581
95% CI	[0.533, 0.635]
Positive recovery fraction	100%

T2 (failed) vs T2.1 v2 (success)

T2: 3/7 valid, recovery = -0.017
(high-temp perturbation, poor calibration)

T2.1 v2: 25/26 valid, recovery = 0.581
(deterministic torsion, 100 kJ/mol/nm²)

Root cause: parameter calibration,
not fundamental method limitation.

Gate pass: 96.2% valid structures, 100% positive contact recovery, mean recovery 0.581

6. 4HIX IDP Antibody Design Validation

Scaffold: 4HIX (humanized 3D6 Fab, Abeta 1-6 DAEFRH). ProteinMPNN T=0.5, 20 samples.

AF2 multimer V3 with VH:VL chain separator. All 20 designs passed validation.

Rank	H3 Sequence	ipTM	pLDDT	iPAE
Native	VRYDHYSGSSDY	0.449	0.194	24.7
#1	LYDESKDAESE	0.462	0.200	24.5
#2	LYDAHHAHSL	0.461	0.192	24.4
#3	LYDGSIGAESQ	0.460	0.195	24.5
#4	LYDSSVDASGH	0.459	0.199	24.7
#5	LFNEANCAESY	0.458	0.199	24.3

Key Findings

- 20/20 designs pass AF2 validation
- Top 3 exceed native ipTM (0.449)
- Best design: ipTM = 0.462 (+2.9%)
- Chain separator fix: VH:VL prevents 4.5x ipTM underestimation

Note: 6-residue epitope; AF2 less discriminative for short peptides. 4HIX in AF2 training set (2012). No wet-lab validation.

7. Evidence Hierarchy and Methodological Rigor

1. ECLS Temporal Final
n=15 clusters, post-2021, sealed

Positive discrimination (0.172, 80%+)

2. ECLS Adaptation
n=46 clusters, exposed

Replicates temporal final pattern

3. T2.1 v2 Recovery
n=31 structures, 96.2% valid

Physical contact recovery (0.581)

4. 4HIX Design
n=20 designs, 20/20 AF2 pass

Prospective design (top ipTM 0.462)

5. T1 Ensemble
n=7 clusters

Single-pose overconfidence (−0.200)

6. Generator Calibration
n=7 clusters, exploratory

Calibrated reranker works

All evaluations use frozen gates, one-shot evaluation on sealed data, and homology-disjoint splitting across 6 axes. Bootstrap CIs with 10,000 resamples.

8. Software Implementation and Reproducibility

Software Components

- BFN core: categorical distributions, focal-weighted losses
- Disorder-augmented dataset: label injection + contrastive profiles
- Receiver network: position-sensitive routing
- IDP design pipeline: 5-stage target-to-validation
- AF2 infrastructure: proper chain separator handling

Reproducibility

- Frozen YAML configuration contracts
- SHA256 manifests for all result files
- results_t2.1_v2_final.json (31 records)
- 4hix_final_validation/final_report.json (20 designs)
- INTEGRATIVE_ANALYSIS.md evidence chain

Critical Bugs Fixed During Development

1. Disorder head: training batches never contained disorder_label → PaddingCollate dropped the field → disorder loss never fired. Fixed with explicit label injection.
2. AF2 chain separator: antibody chains must use VH:VL (not concatenation) or ipTM drops 4.5x.
3. T2.1 perturbation: peptide restraint 25 → 100 kJ/mol/nm² improved valid fraction 43% → 96.2%.

9. Conclusions and Future Work

Conclusions

- ECLS establishes epitope-conditioned CDR-H3 signal (46 clusters, mean 0.217, 80.4% positive)
- Temporal transfer confirmed on sealed post-2021 data (mean 0.172, 80% positive, $P=0.014$)
- T2.1 v2: 96.2% valid structures, 100% positive contact recovery (0.581)
- 4HIX prospective design: 20/20 pass AF2, top 3 exceed native ipTM
- Disorder-aware framework provides principled approach to IDP antibody design

Limitations

- No wet-lab validation; all results computational
- Short epitope (6 residues); AF2 less discriminative; 4HIX in AF2 training set

Future Work

- CAID benchmark evaluation of retrained disorder head
- Extended IDP targets (tau, alpha-synuclein) and experimental validation (SPR/ELISA)